

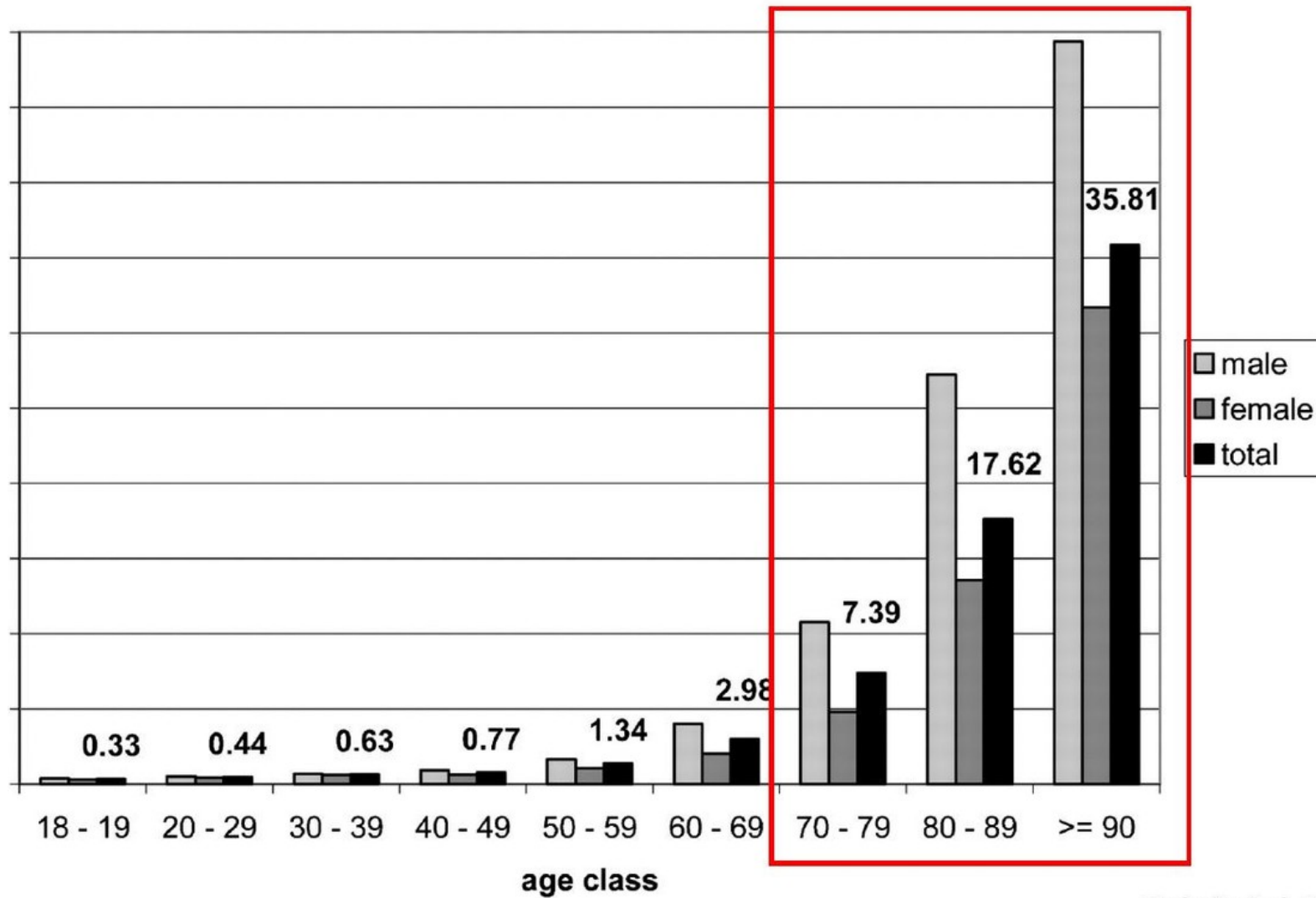
Lecture 1 — Respiratory Tract Infections & Immunity

Companion Figures

Dr James Harker · CVR module

Companion figures referenced by ■ markers in the Notion revision page. Figures ordered as they appear in the .md. Supplementary figures listed at the end for completeness.

pneumonia rates increase with age



Ewig S et al. Thorax 2000;64:1062-1068

Figure L1.5 — Hospitalised CAP incidence per 1,000 by age class (Ewig 2009)

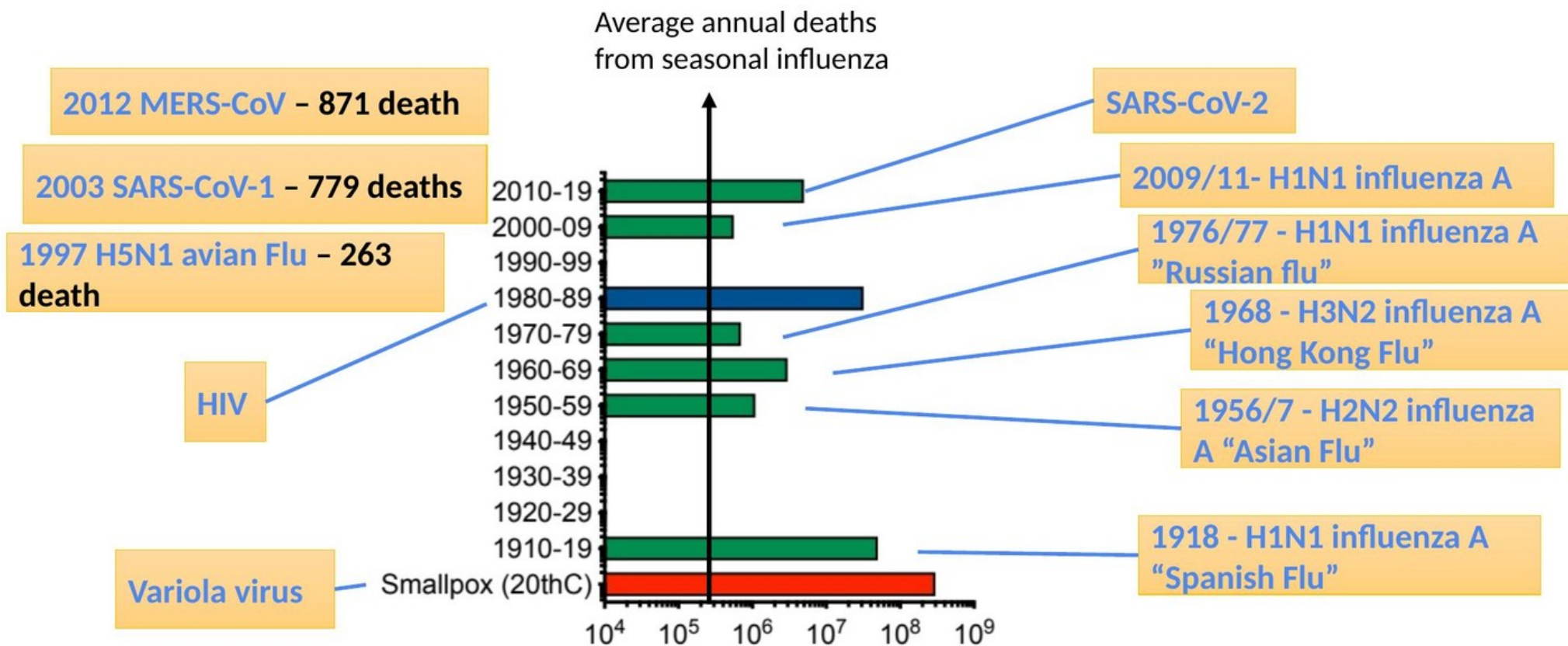


Figure L1.9 — Average annual deaths from seasonal influenza with historical pandemics labelled

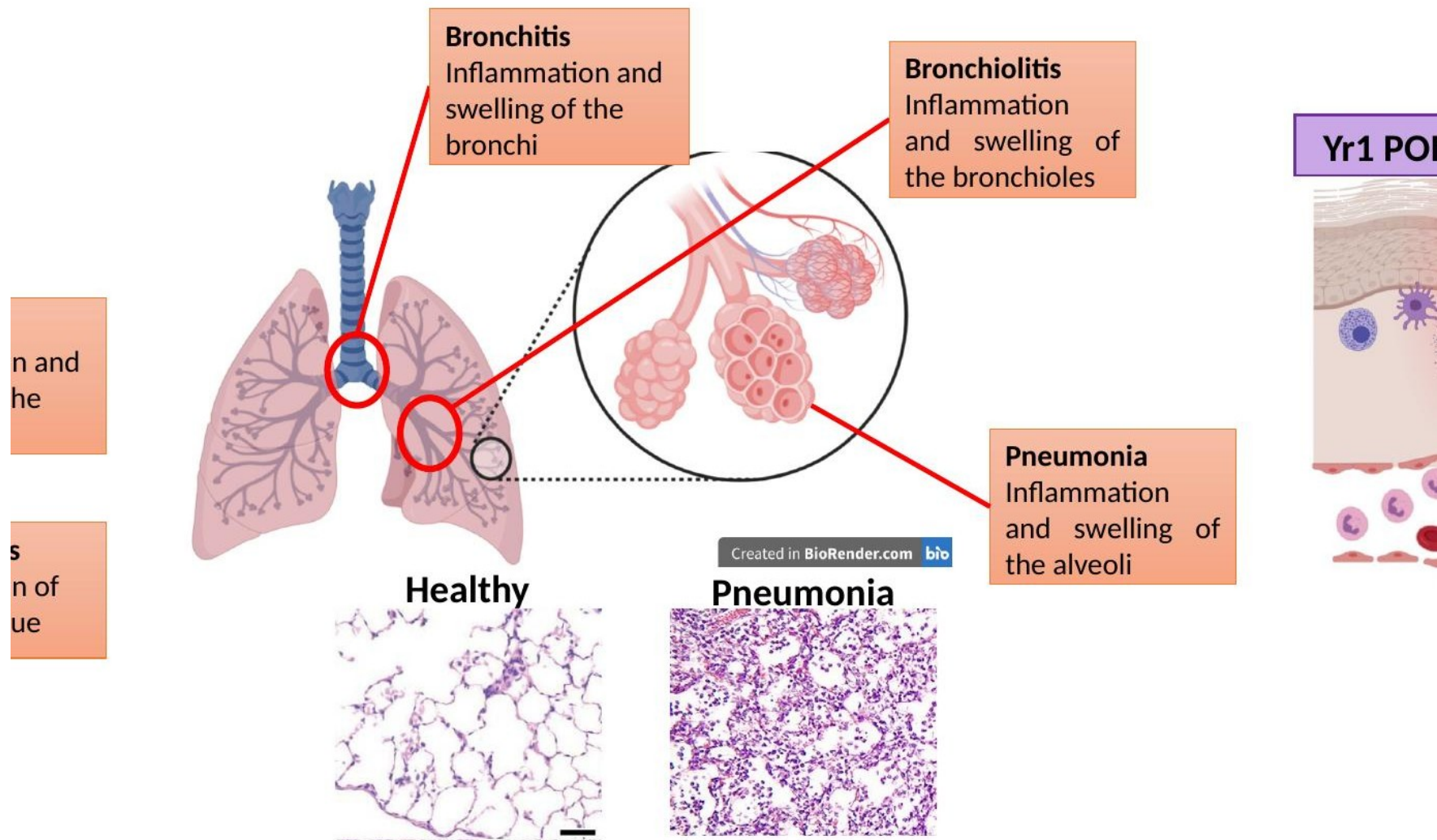


Figure L1.10 — Anatomical map of respiratory tract inflammations and alveolar histology

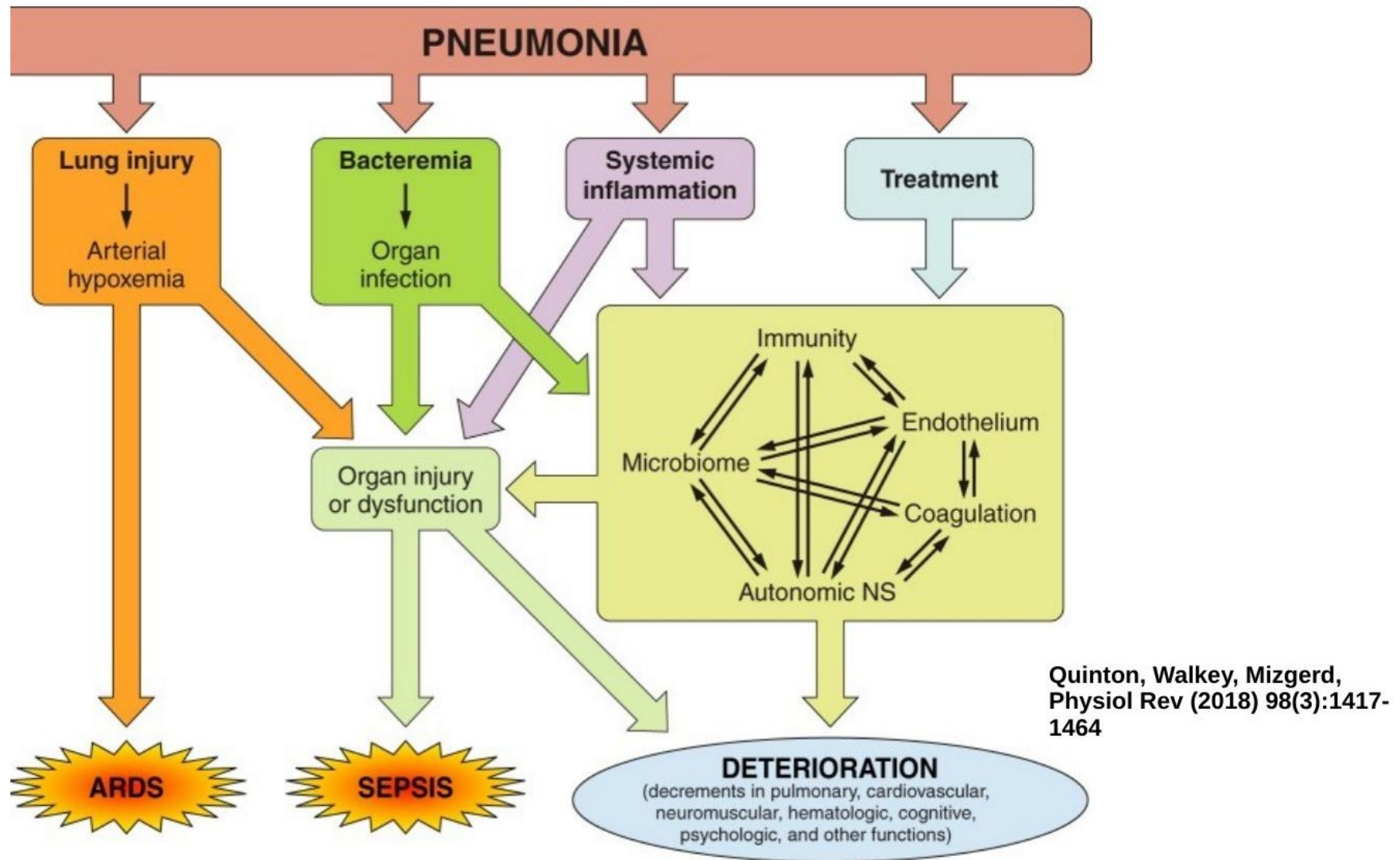


Figure L1.11 — Downstream consequences of bacterial pneumonia (Quinton 2018)

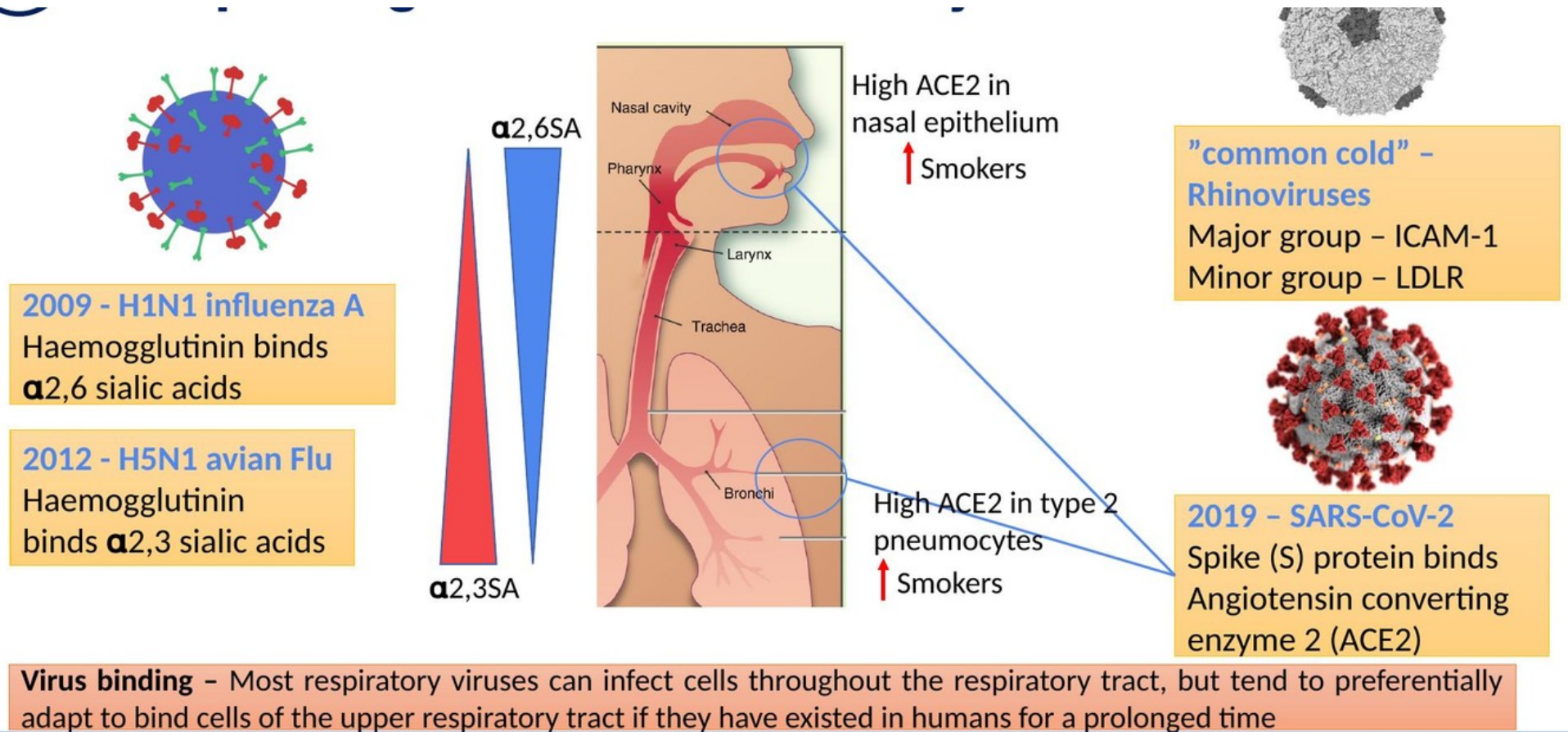
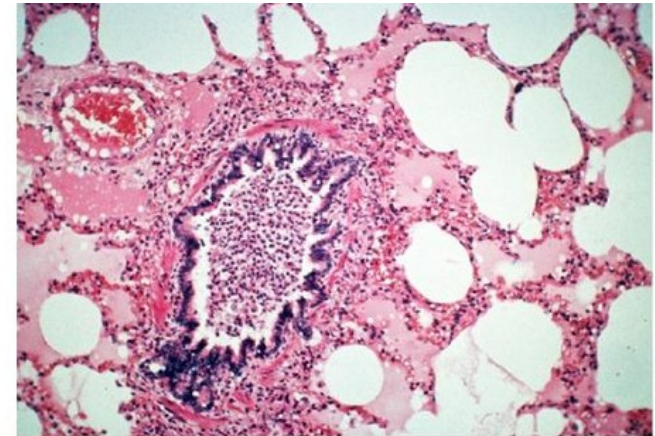
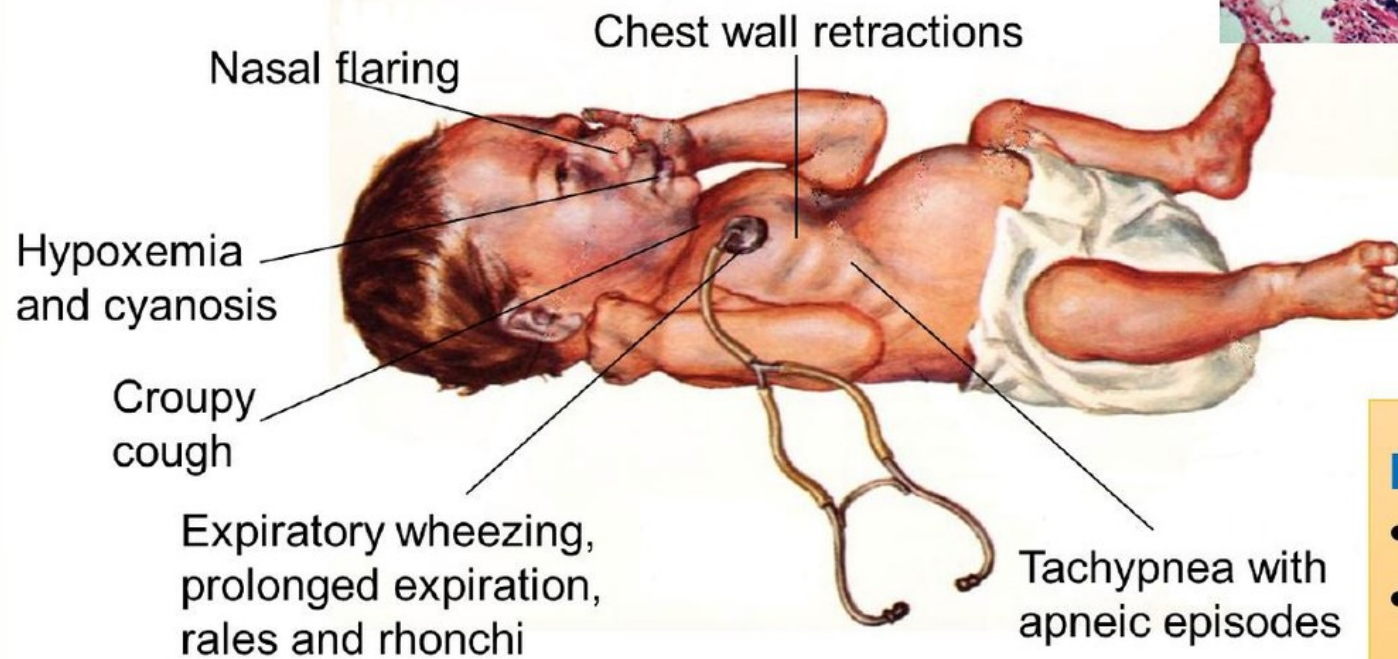


Figure L1.14 — Viral receptor tropism: sialic acid distribution and ACE2 in the respiratory tract

itis in infants - RSV

d



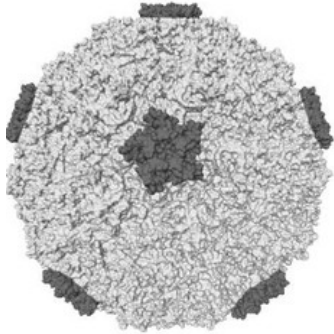
Risk factors

- Premature birth
- Congenital heart and lung disease

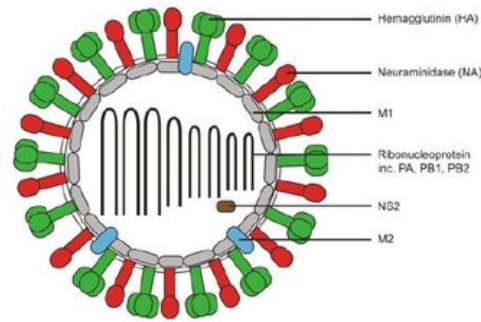
Cochrane Database Syst Rev 2000; CD001266

Figure L1.15 — Clinical signs of severe RSV bronchiolitis in infants

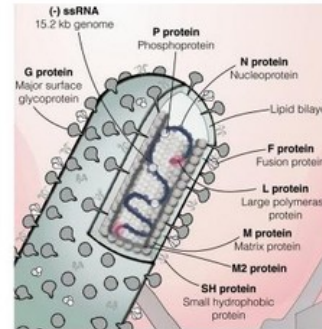
Rhinovirus



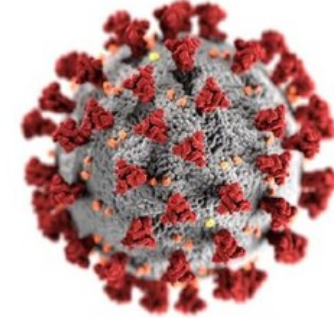
Influenza



RSV



SARS-CoV-2

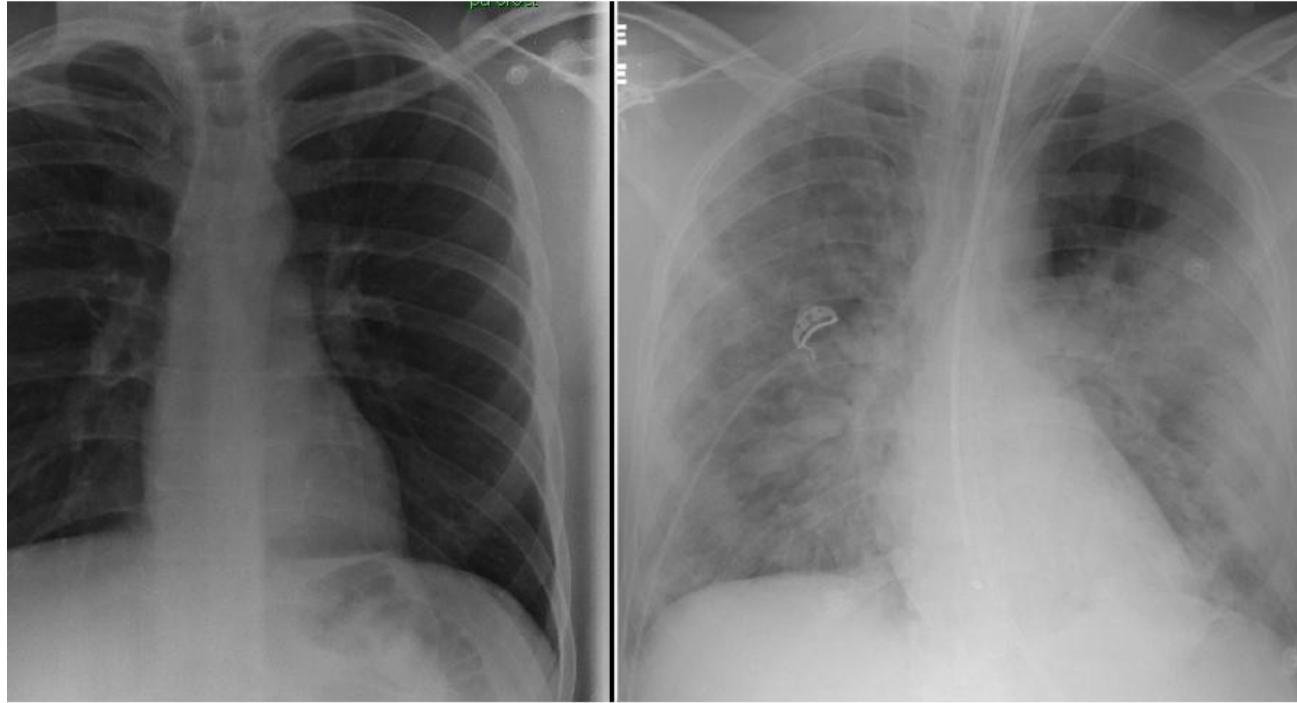


• Antigenically novel

Figure L1.20 — Comparative surface antigen biology: rhinovirus, influenza, RSV, SARS-CoV-2

MILD

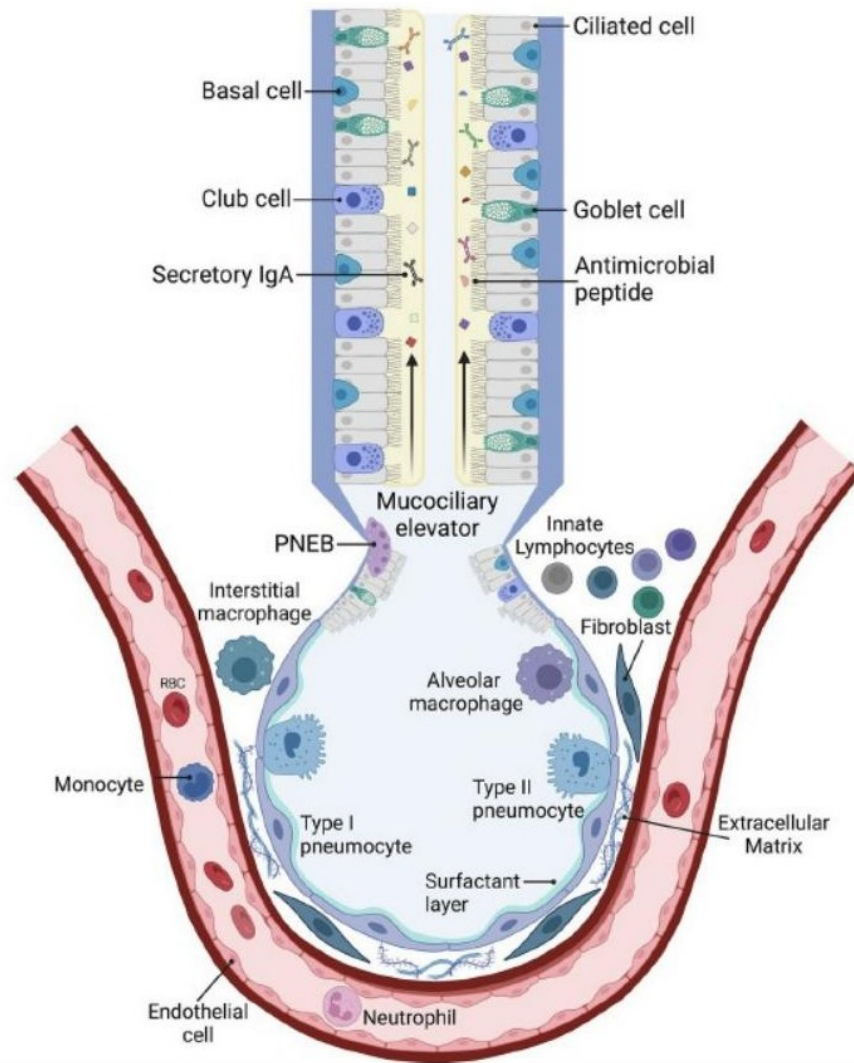
SEVERE



same viral strain:
pandemic H1N1 ('swine flu') 2009-10

Why are they different?

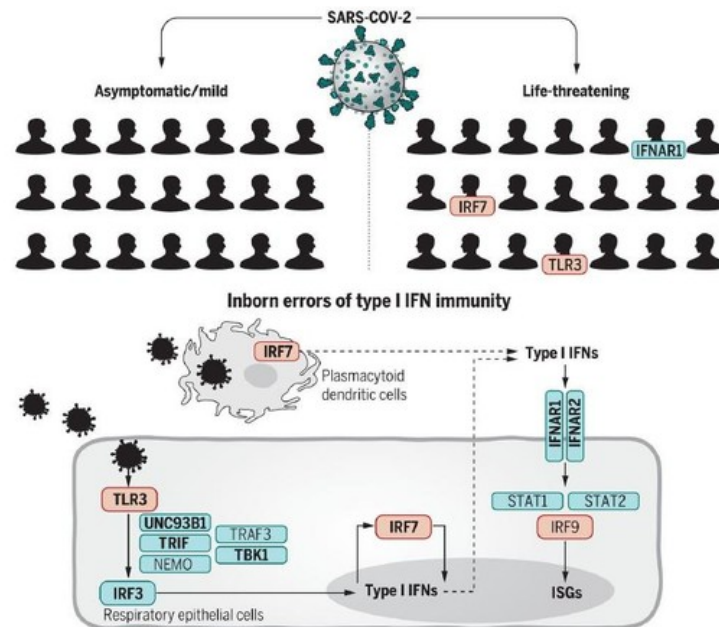
Figure L1.16 — Differential outcomes in pandemic H1N1 2009: chest X-rays



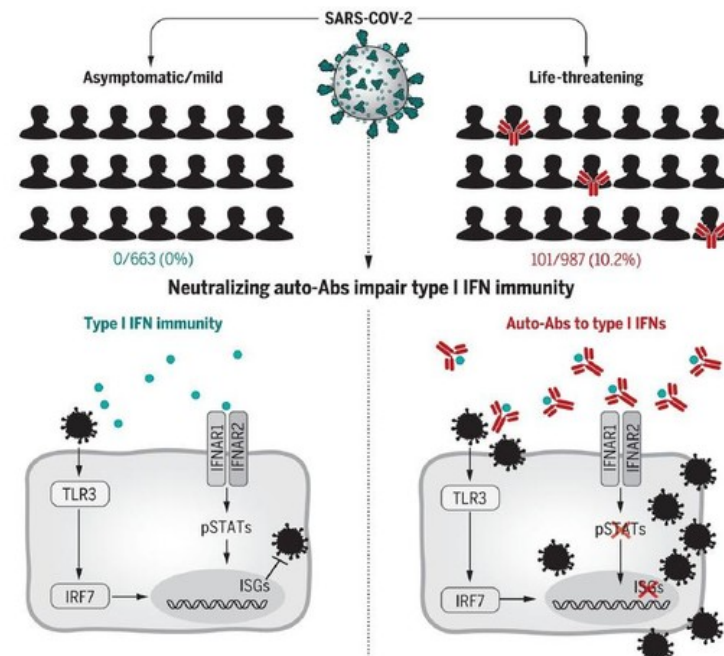
Res

- T
- S
- M
- C
- A
- r
- n
- F
- C
- I
- b
- u
- a

Figure L1.17 — Cellular architecture of the respiratory epithelium (Invernizzi 2020)



Zhang et al, Science 2020



Bastard et al, Science, 2020

Observation – While this individuals had life threatening SARS-CoV-2, they did not have a history of severe respiratory

Figure L1.18 — Type I interferon pathway: inborn and acquired defects in severe SARS-CoV-2

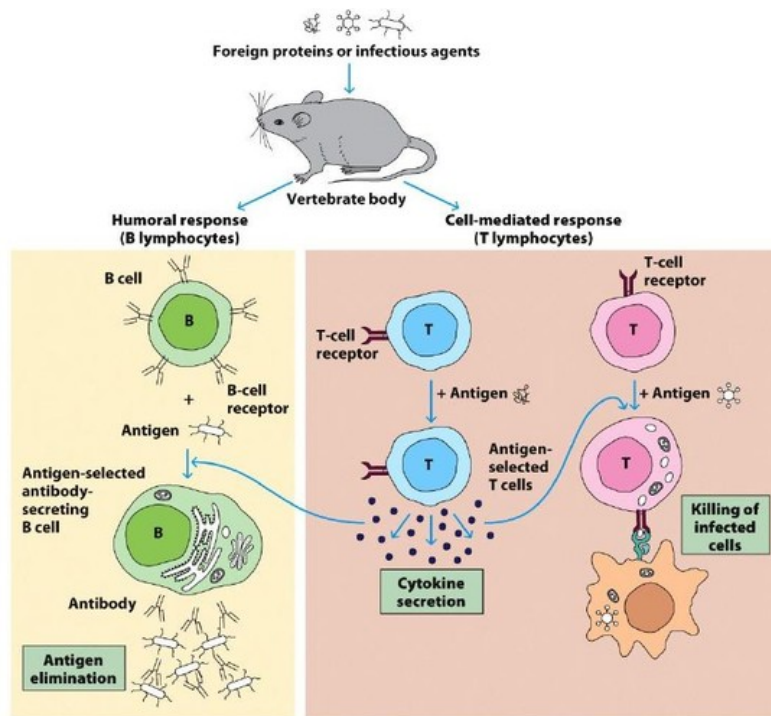


Figure 1-5
Kuby Immunology, Seventh Edition
© 2013 W. H. Freeman and Company

- first infection.
- **CD8 T cells** provide antigen specific cytotoxic immunity and immunological memory (both resident and systemic).
 - **B cells** differentiate into antigen specific antibody secreting cells and memory B cells. Providing neutralizing immunity.

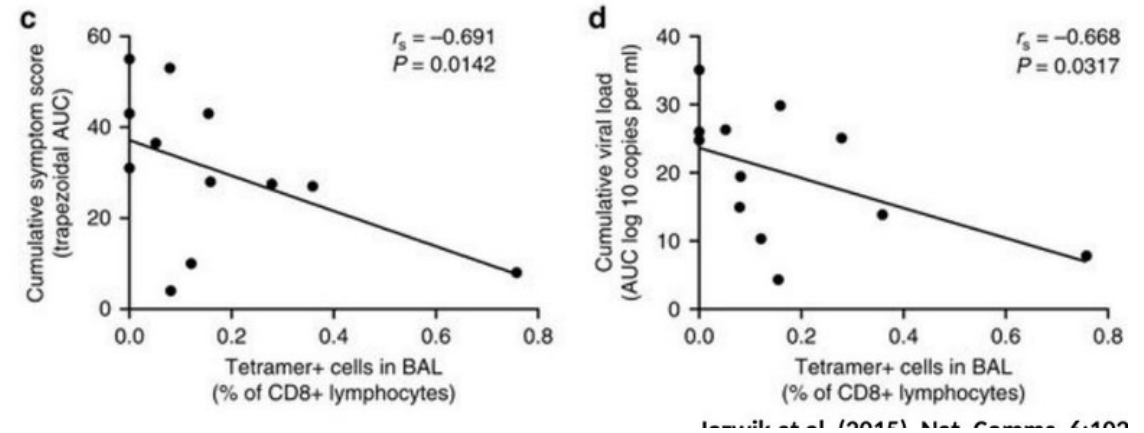


Figure L1.19 — Adaptive immunity and CD8 tetramer cells vs symptom and viral load (Jozwik 2015)

In 1919, Abrahams and colleagues³² reviewed their extensive experience during the 1918–19 influenza pandemic. They reported that *Streptococcus pyogenes longus* (36%), pneumococcus (29%), and *B influenzae* (25%) were recovered relatively frequently from cultures (n=28) of lung tissue of fatal “influenzal pneumonia”

Expeditionary Forces in Europe documented a sharp rise in reports of cerebrospinal meningitis approximately 1 week after sharp increases in reports of influenza and pneumonia. The bulletin noted that “it has been a usual observation that when infections of the upper respiratory tract prevail, the incidence of meningitis in the community increases soon after and this rule prevails at present”.⁶⁷



Figure 2: Faces of “influenzo-pneumonic septicaemia”

(A) “An early case in which the facial colour is frankly red, and the patient might not appear ill were it not for the drooping of the upper eye-lids and a half-closed appearance to the eyes.” (B) “Cyanosis in which the colour of the lips and ears arrests attention in contrast to the relative pallor of the face. The patient may yet live for twelve hours or more.” (C) “The heliotrope cyanosis. The patient is not in physical distress, but the prognosis is almost hopeless.” Reproduced from reference 32.

Lancet Infect Dis 2006; 6:

Figure L1.22 — Faces of influenzo-pneumonic septicaemia, 1918–19

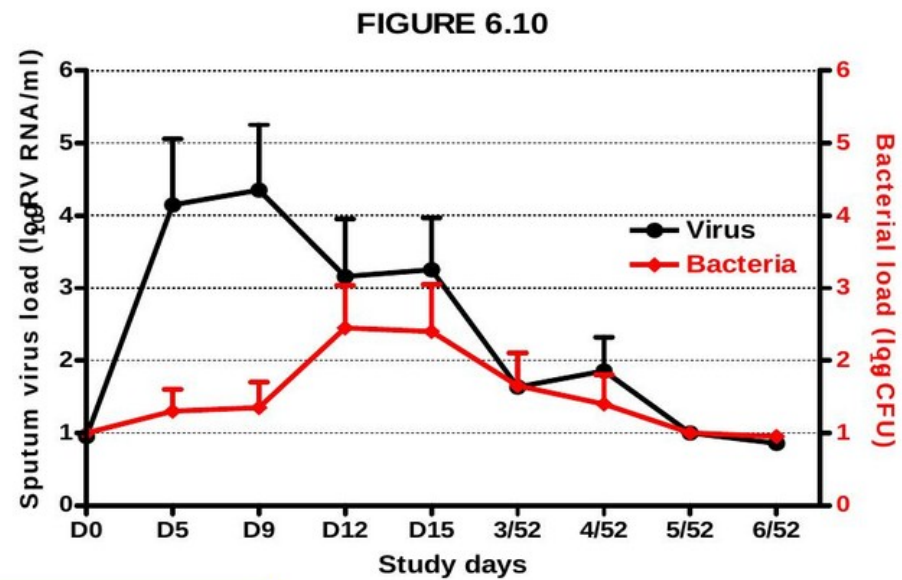
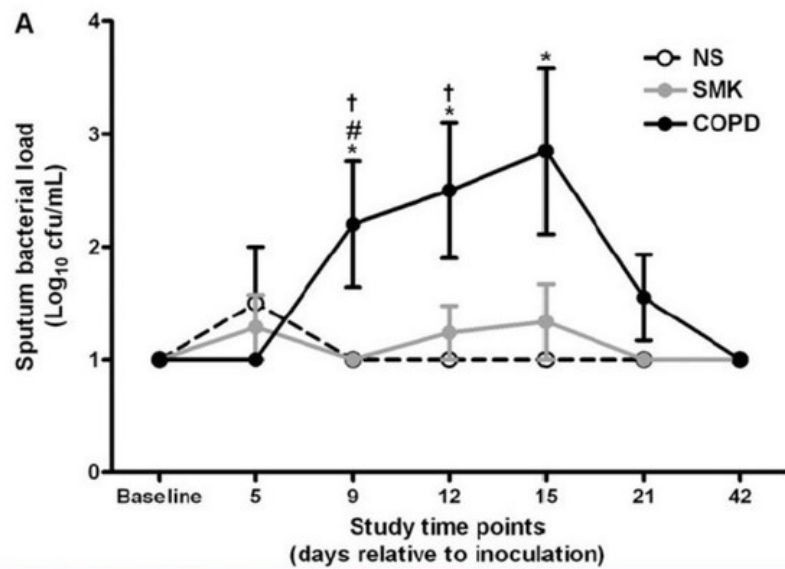
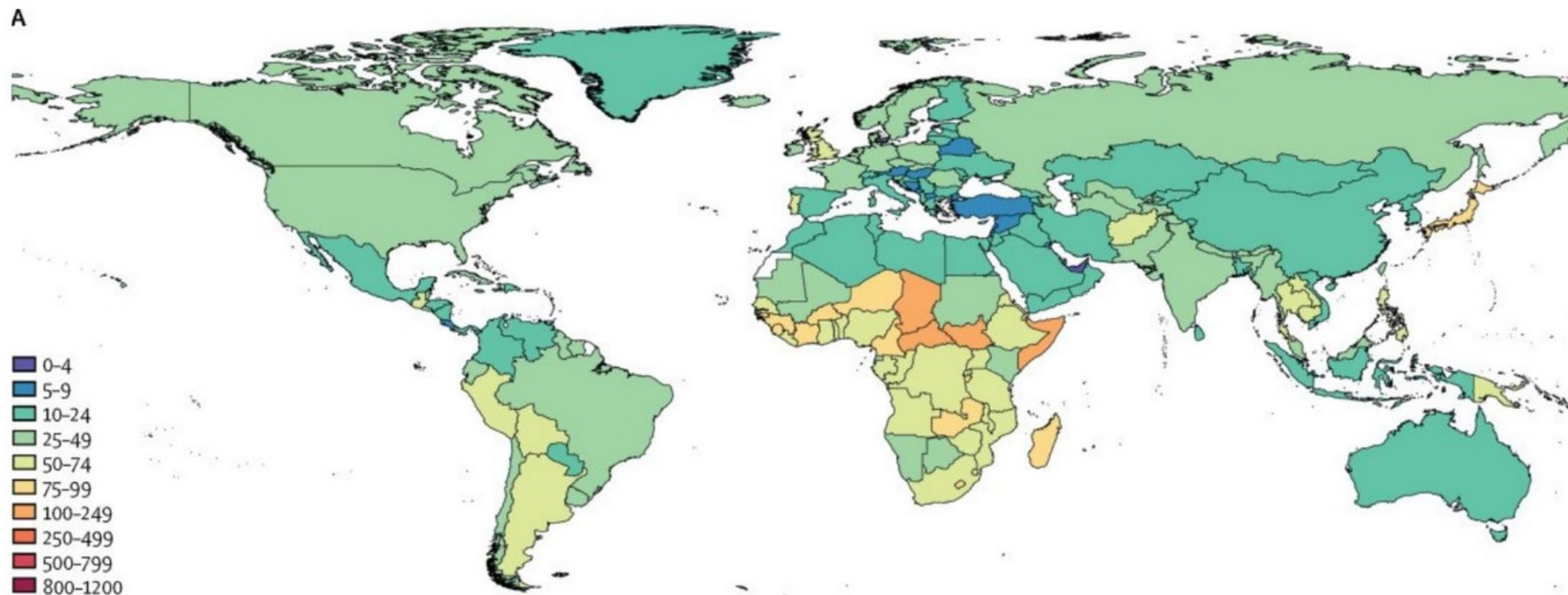


Figure L1.23 — Rhinovirus-induced secondary bacterial infection in COPD (Mallia 2012)

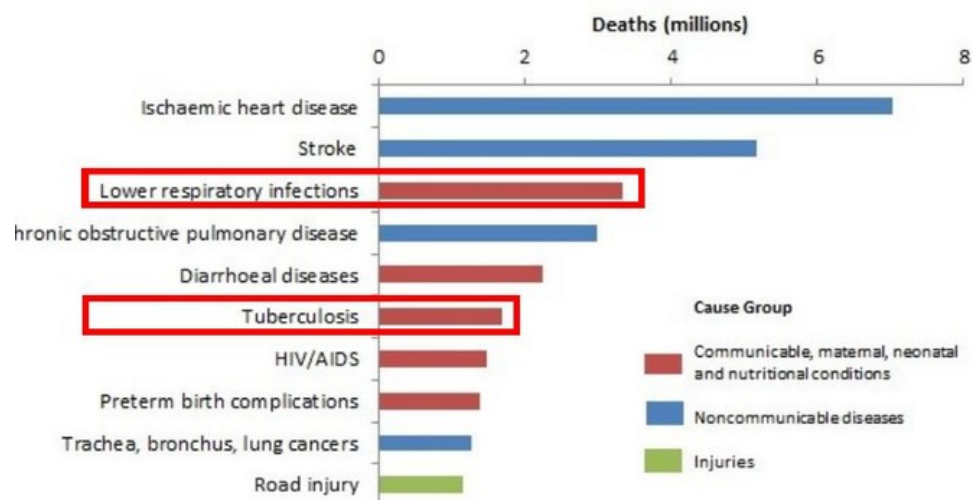


LRI mortality rate per 100,000 of the population

Lancet Infect Di

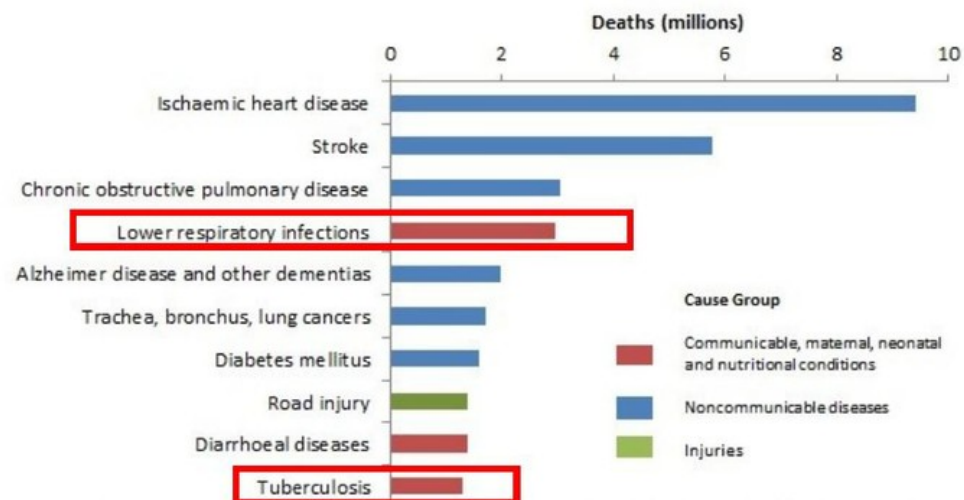
Figure L1.1 — Global LRI mortality rate per 100,000 (2015) [supplementary]

Top 10 global causes of deaths, 2000



Source: Global Health Estimates 2016: Deaths by Cause, Age, Sex, by Country and by Region, 2000-2016. Geneva, World Health Organization; 2018.

Top 10 global causes of deaths, 2016



Source: Global Health Estimates 2016: Deaths by Cause, Age, Sex, by Country and by Region, 2000-2016. Geneva, World Health Organization; 2018.

~5.2 million deaths in 2000



~4 million deaths in 2016

Figure L1.2 — Top 10 global causes of death (2000 vs 2016), with LRTI and TB highlighted [supplementary]

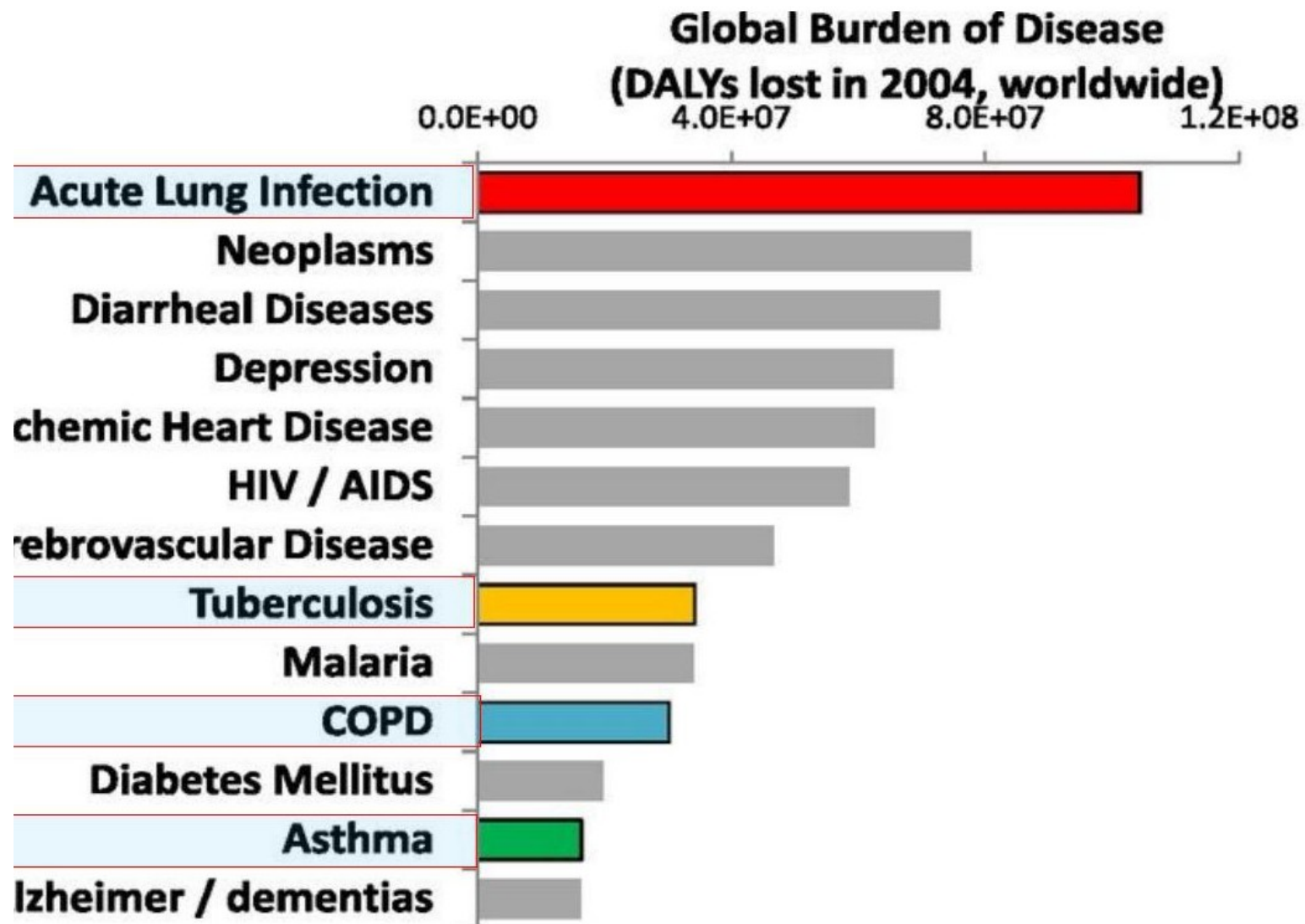
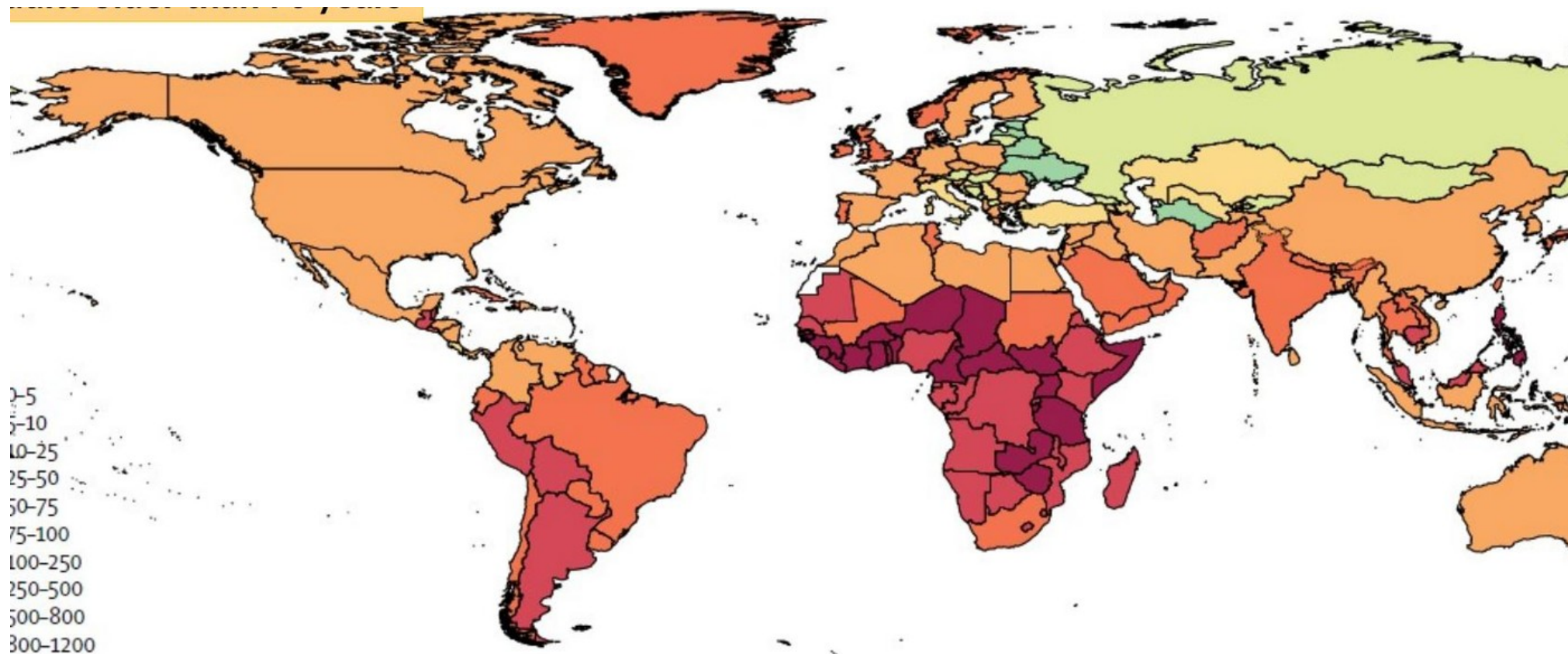


Figure L1.3 — Global burden of disease by DALYs (2004) [supplementary]

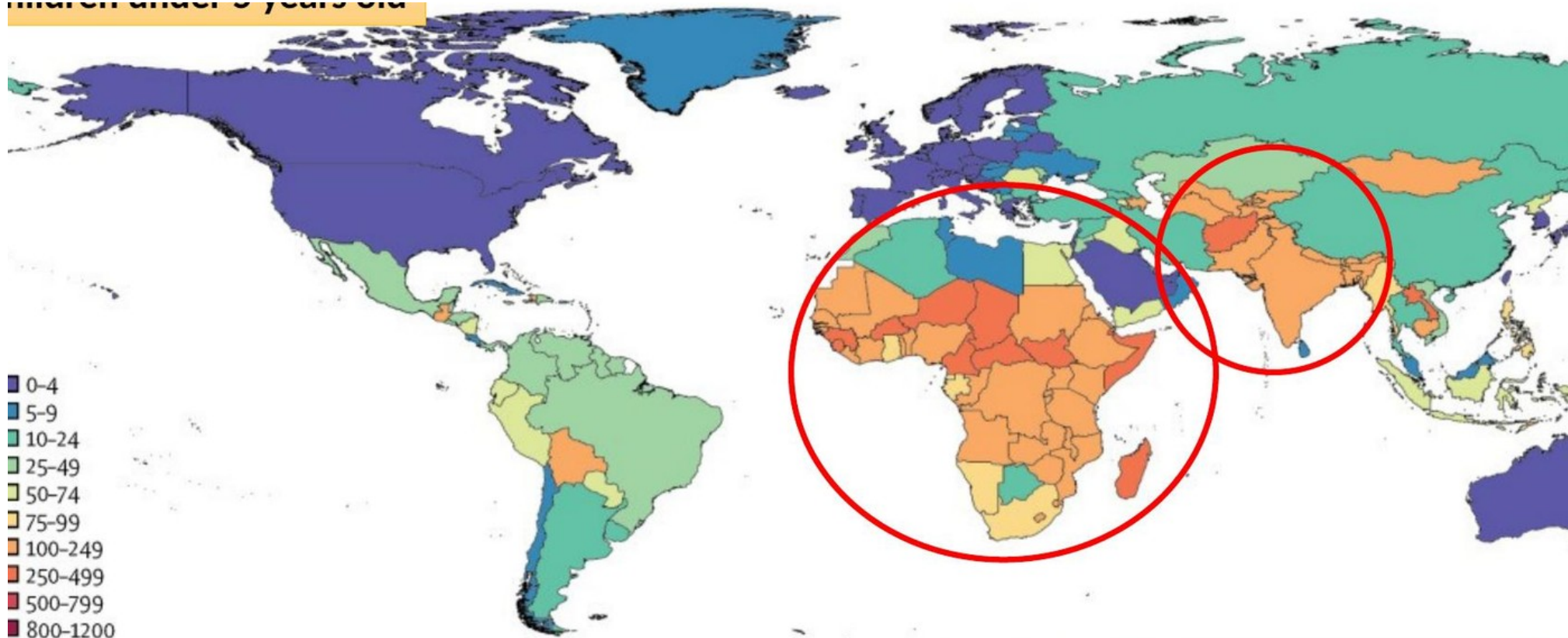


LRI mortality rate per 100,000 of the population

Lancet Infect Di

Figure L1.4 — LRI mortality rate per 100,000 — adults >70 [supplementary]

children under 5 years old



LRI mortality rate per 100,000 of the population

Lancet Infect Di

Figure L1.6 — LRI mortality rate per 100,000 — children <5 [supplementary]

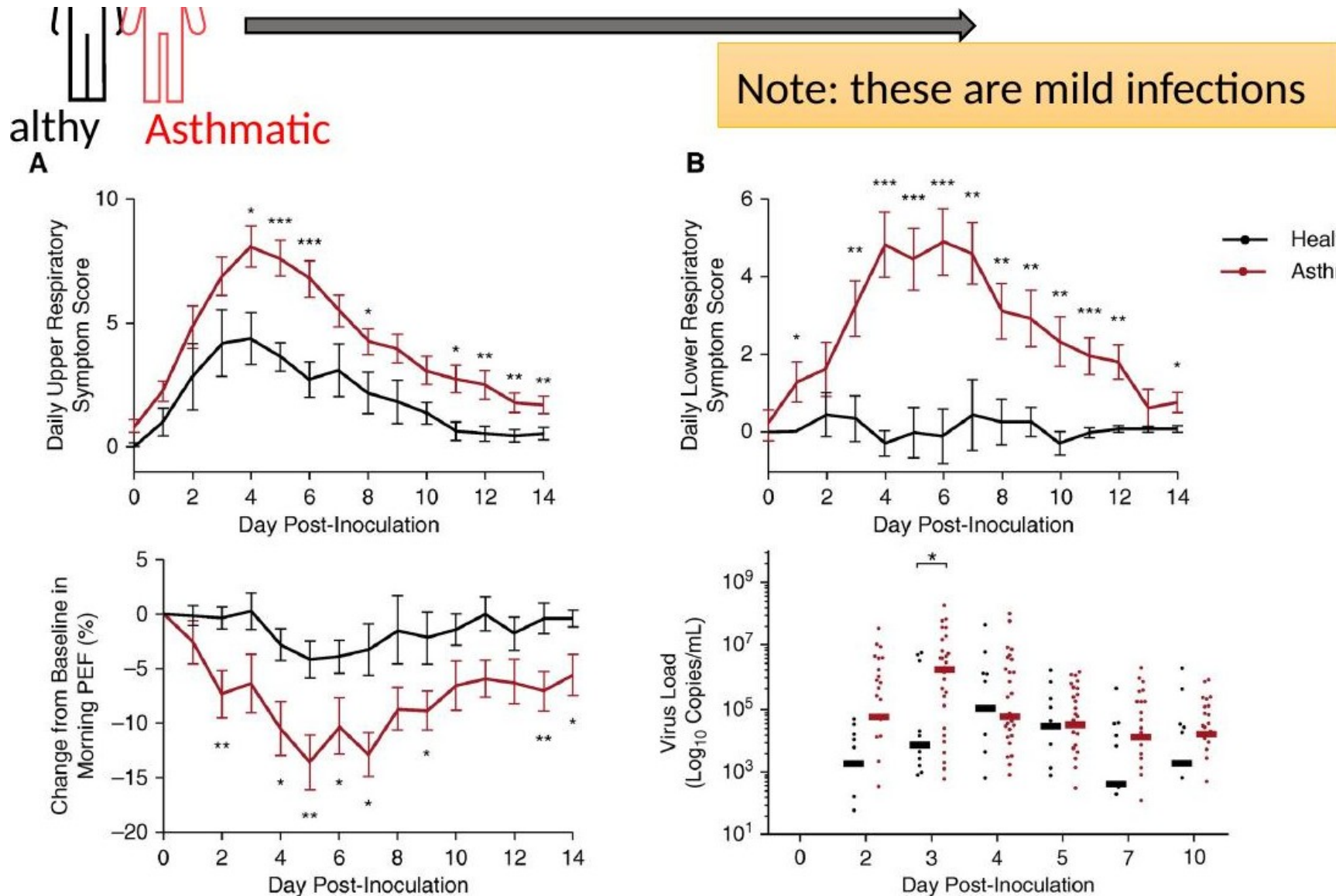
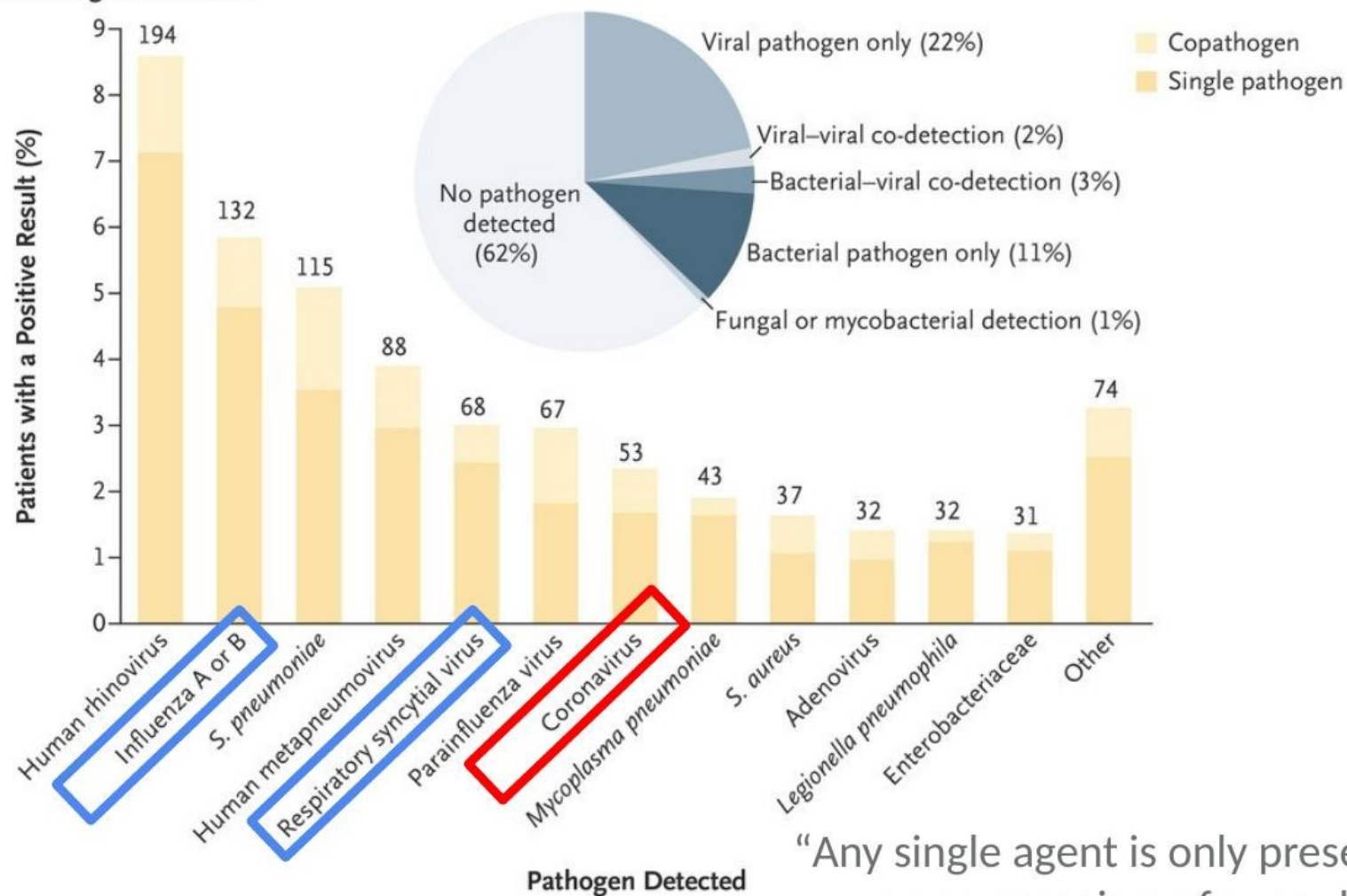


Figure L1.7 — Experimental rhinovirus challenge: healthy vs asthmatic (Jackson 2014) [supplementary]

Specific Pathogens Detected



“Any single agent is only present in a minor on an organism of overwhelming impor

Figure L1.8 — Pathogens identified in CAP (Jain 2015) [supplementary]

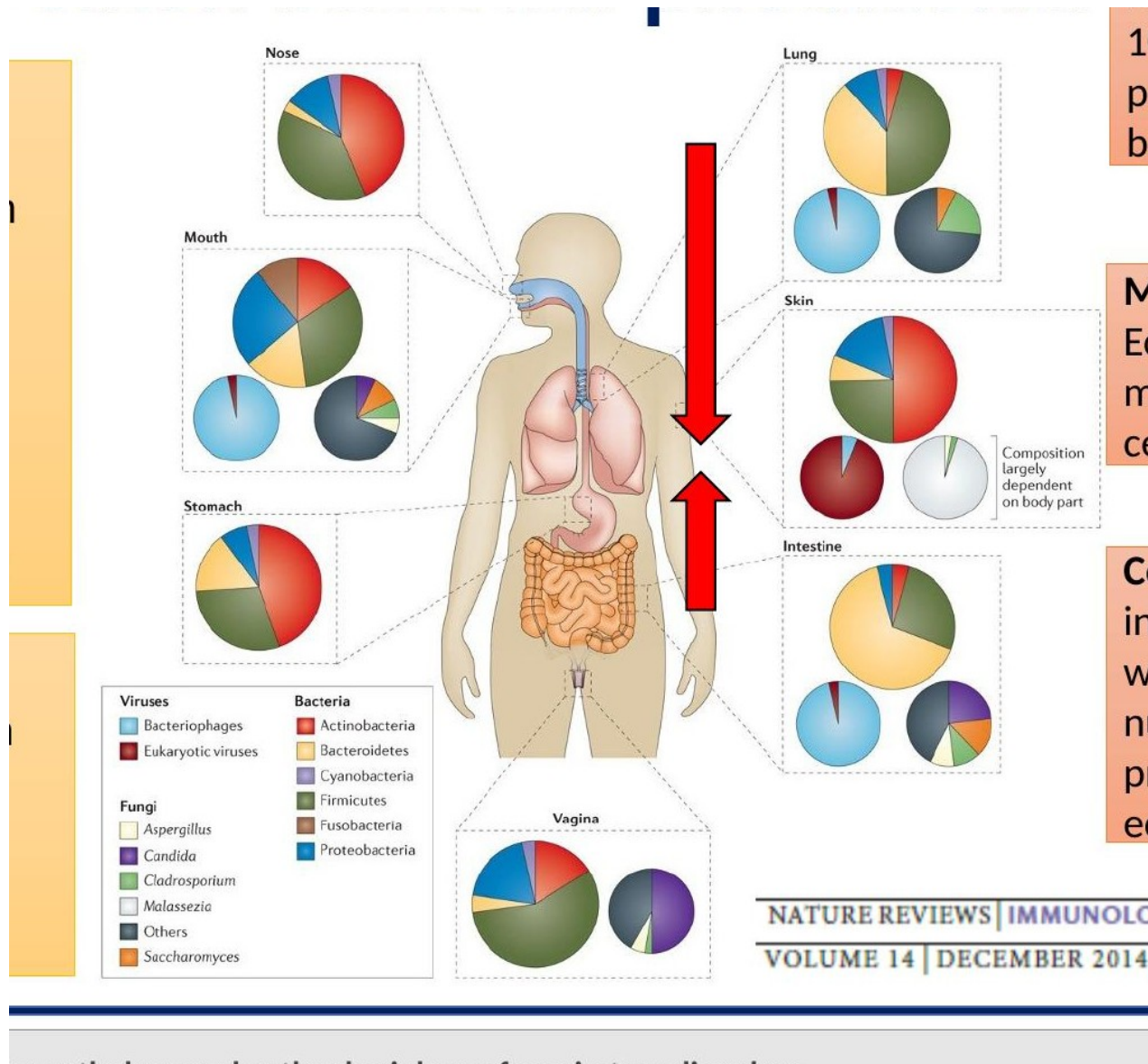
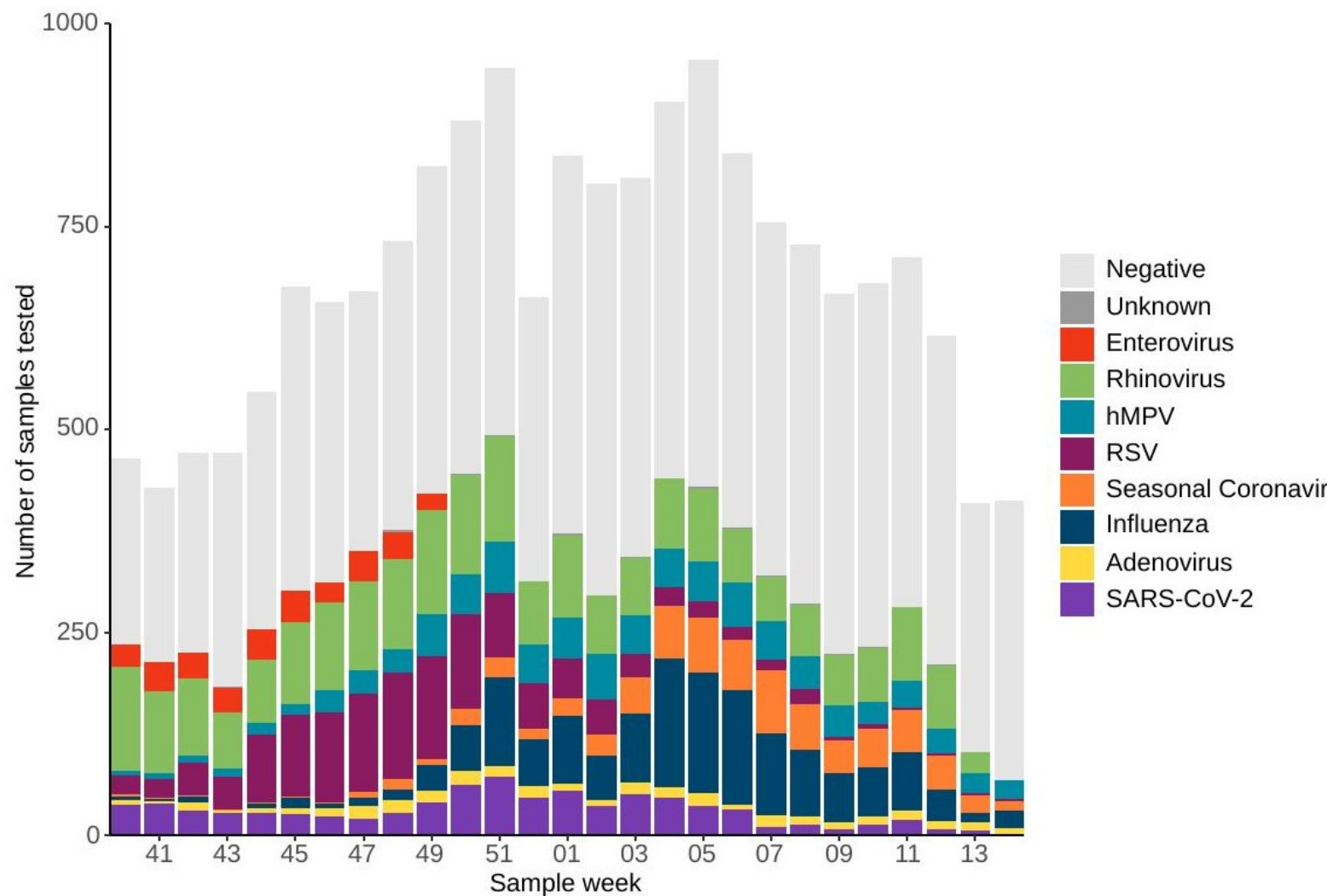


Figure L1.12 — Human microbiome composition across body sites [supplementary]

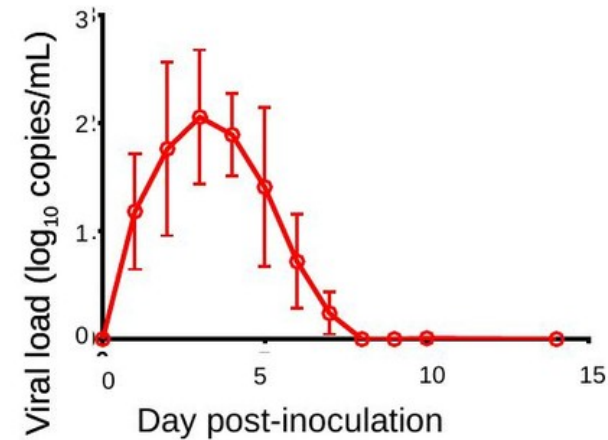
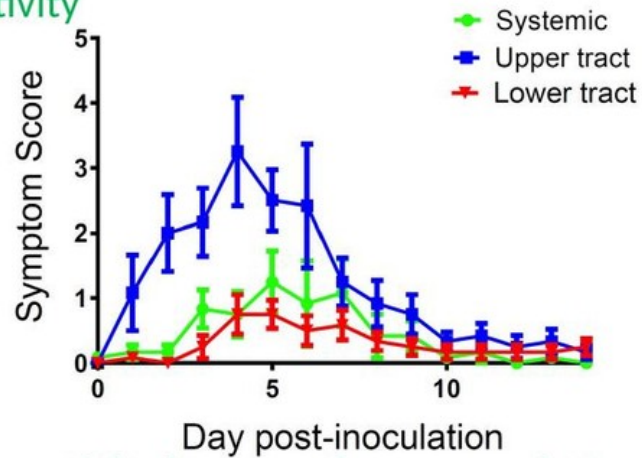


2023-2024 English GP swab surveillance (gov.uk)

Figure L1.13 — UK GP swab virus surveillance 2023–24 [supplementary]

seronegativity

enza



l for seronegativity because there are only 2
e've had both by 3 years old

SV

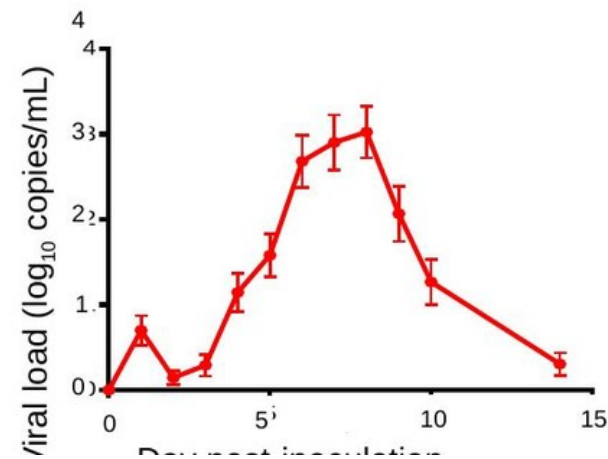
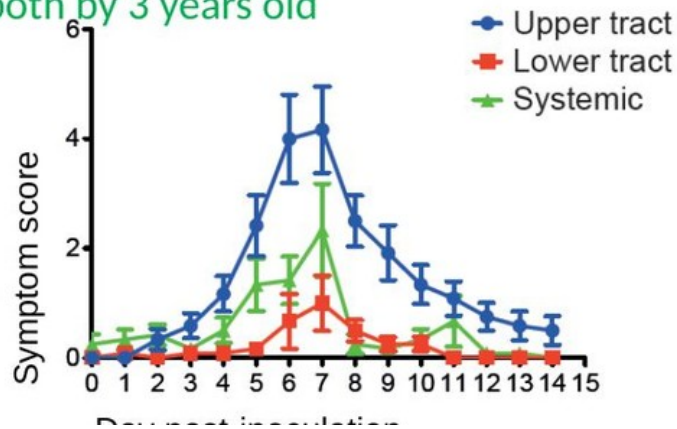


Figure L1.21 — Symptom and viral load kinetics: influenza vs RSV human challenge [supplementary]